

IEEE 14-Bus with Controller

Production 160-s Comparison of the Legacy Selector, ET-FCSPS, and Finite-Set BO Replay

In-house Power-Flow Project

4 August 2026

Abstract

Three controller arms were evaluated on the same project-owned all-KCL IEEE 14-bus simulation. Case data, initial state, SG/IBR equations, event chronology, dispatch, synchronizer, solver, tolerances, and the fixed 0.0125-s step were identical. The legacy selector, exhaustive event-triggered finite-control-set predictive supervisor (ET-FCSPS), and finite-set Bayesian-optimisation (BO) replay all selected IBR1–IBR4 as GFM with IBR1 as the island reference owner. All runs reached 160 s and their raw trajectories were numerically identical. In this four-IBR case ET-FCSPS is preferable: only eight candidates pass the authenticated pre-screen, while the frozen BO budget is also eight, so BO provides no evaluation reduction or response improvement.

1 Controller contract

AGSI++ and authenticated events trigger an evaluation; they do not directly select the mode vector. Every admissible candidate starts from the same accepted event-left state and receives the same SG-trip transaction, device-owned state transfer, post-trip dispatch, full-KCL right-limit solve, and $T_p = 0.25$ s production-kernel prediction. Hard KCL, voltage, frequency, RoCoF, current, reserve, dwell, lockout, online-resource, and reference-continuity gates precede the cost

$$J = \sum_k w_k \phi_k. \quad (1)$$

The terms cover voltage, frequency, RoCoF, P/Q reserve, current utilisation, handback mismatch, GFM count, and switching count. Weights and normalisers were frozen before closed-loop results.

The SG uses the operational sixth-order EMF state

$$\mathbf{x}_{SG} = [\delta, \Delta\omega, E'_q, E'_d, E''_q, E''_d]^T, \quad 2H\dot{\Delta\omega} = P_m - P_e - D\Delta\omega. \quad (2)$$

Each dual-mode IBR retains a 13-state GFM subset and a 7-state GFL subset. The supervisor changes only the discrete active subset and reference-owner metadata; it does not modify device equations.

2 BO comparator

BO is an **offline finite-set replay** over the identical immutable trial table. It uses an in-house RBF Gaussian process and Expected Improvement,

$$k(\mathbf{z}, \mathbf{z}') = \exp\left(-\frac{\|\mathbf{z} - \mathbf{z}'\|^2}{2\ell^2}\right), \quad (3)$$

with a frozen budget of eight. It does not call an external optimiser and is not presented as an online production authority.

3 Common 160-s experiment

The chronology is SG trip at 20 s, 20% load increase at 50 s, bus-9 fault from 85 to 85.15 s, line 6–13 trip at 110 s, and restoration/reclose request at 145 s. The SG passes its synchronism guard at 147.175 s, after which the coordinated reference handback returns all IBRs to GFL.

Table 1: Raw production results

Metric	Legacy	ET-FCSPS	BO replay
End time (s)	160.000	160.000	160.000
Minimum voltage including fault (pu)	0.060077	0.060077	0.060077
Maximum voltage (pu)	1.133568	1.133568	1.133568
Online frequency range (Hz)	58.3238–63.5606	58.3238–63.5606	58.3238–63.5606
Maximum current utilisation	1.0000	1.0000	1.0000
Total mode switches	8	8	8
Aggregate GFM-on time (s)	508.700	508.700	508.700
SG reclose/handback (s)	147.175	147.175	147.175
Maximum accepted residual	9.982×10^{-9}	9.982×10^{-9}	9.982×10^{-9}
Prediction evaluations	–	8	8
Selected GFM set	IBR1–IBR4	IBR1–IBR4	IBR1–IBR4

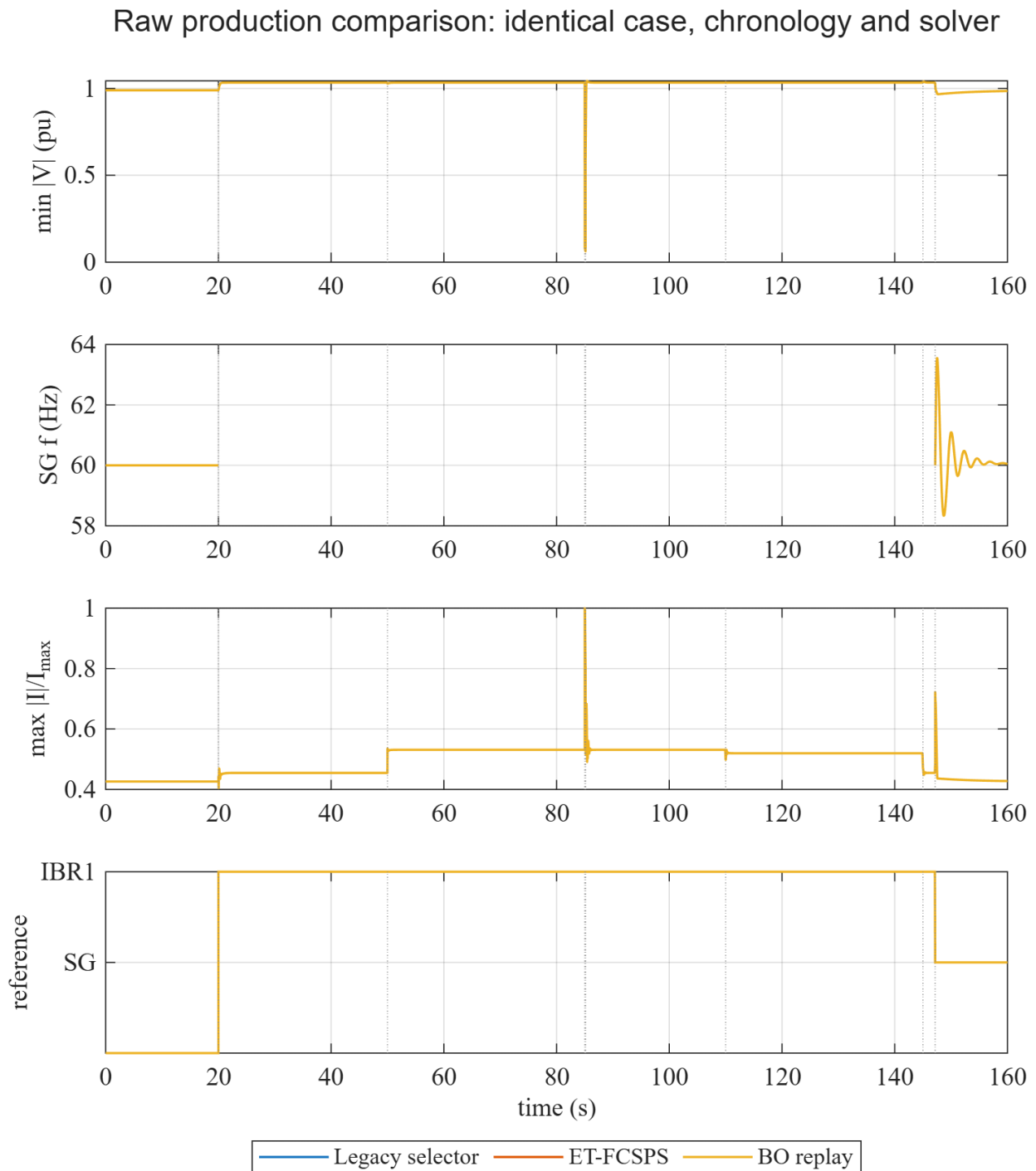


Figure 1: Raw system responses. The three lines coincide because all methods commit the same action.

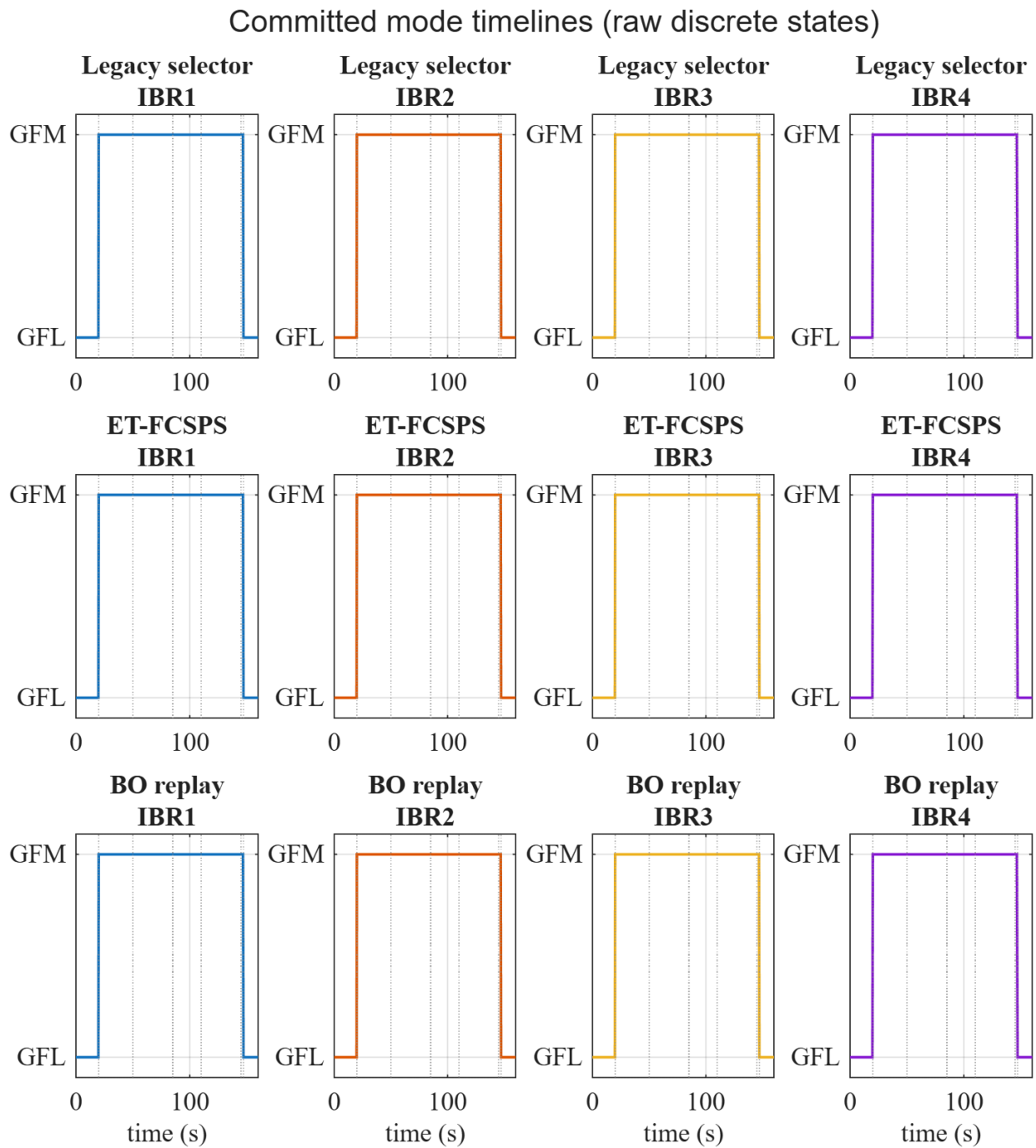


Figure 2: Committed mode timelines for all four IBRs and all three methods.

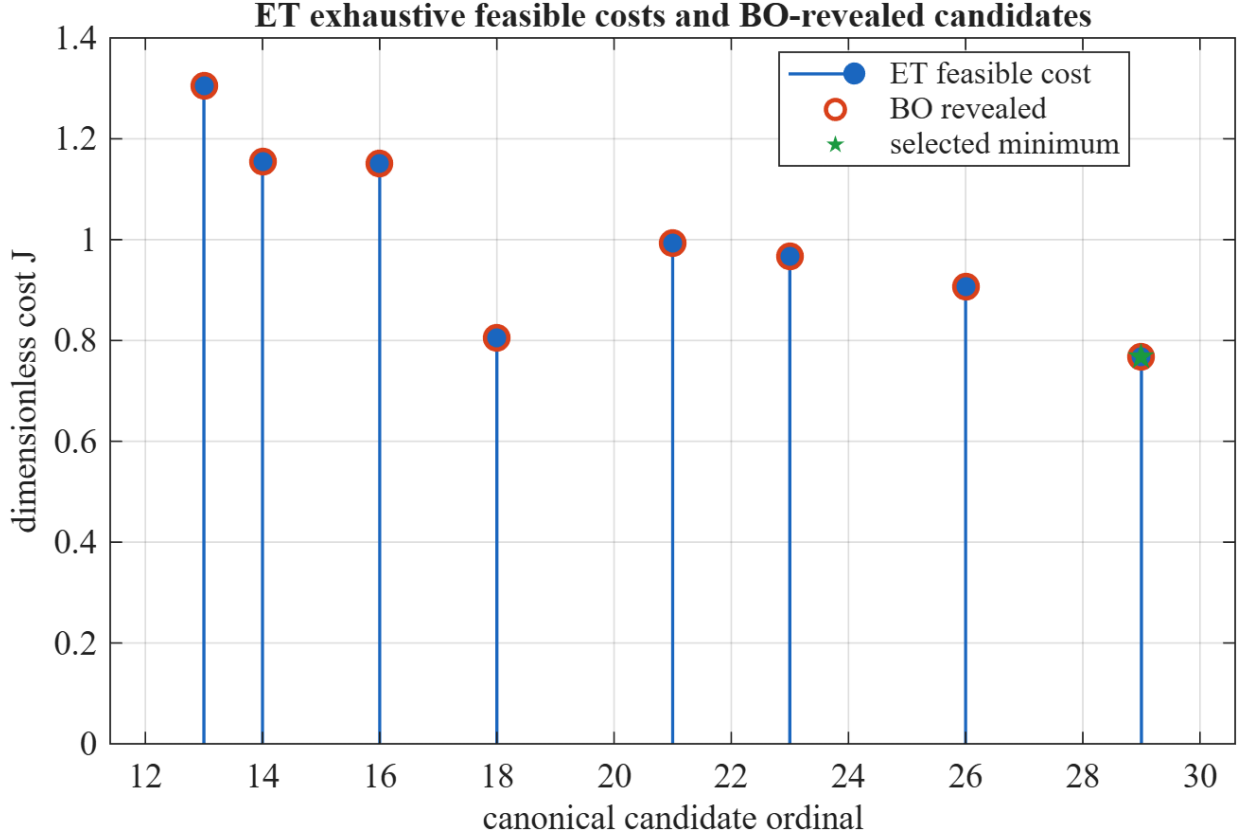


Figure 3: Eight feasible ET costs and the eight BO-revealed candidates. The BO budget spans the full feasible set.

4 Interpretation

The maximum absolute ET-versus-legacy, BO-versus-legacy, and BO-versus-ET differences are exactly zero for AGSI++, discrete mode, P, Q, i_d , i_q , frequency, angle, bus voltage, network minimum voltage, reference code, and SG signals. This is expected: the selected discrete action and continuous event-right state are identical, and the production solver is deterministic.

For this case ET-FCSPS is the stronger method. Exhaustive evaluation is tractable, gives a deterministic finite-set global minimum, and introduces no surrogate hyperparameters. BO cannot reduce work because its budget of eight equals the feasible-candidate count. BO may become useful for a larger IBR fleet, but that requires a separate test with a budget below the feasible-set size and measured regret.

The ET and BO long-run wall times were 2182.918 s and 2223.825 s. Their 1.9% difference is ordinary runtime variation, not a controller benefit, because the dynamic trajectories are identical and common candidate evidence was reused. The historical baseline-cache runtime is not used as a speed benchmark.

5 Limitations and reproduction

P/Q reserve uses the existing case-defined nameplate proxy; energy duration and hardware reserve are not established. BO remains offline replay. The result validates this software/controller chronology, not protection, HIL, or field readiness. No equation, threshold, weight, event, solver setting, or raw plot was tuned after viewing the result.

```
pf_init_paths;  
addpath(fullfile('scripts','reporting'));  
out = run_ieee14_controller_comparison(reuse_completed=true);  
generate_ieee14_controller_comparison_figures;
```